Lectures

MATH 107: Introduction to Linear Algebra — Fall 2026

Daily reading is listed with each meeting. Free Strang/MIT files and the Lachniet Octave PDF are collected on Resources.

Weeks 1–4 outlines are posted. Lectures through Week 4 stay one class day later than the original calendar (Lecture 01 began Wednesday, August 26). **Week 5 order:** dependence → span → basis + Quiz 4. **Week 6:** neighbors practice (Mon) → Midterm 1 prep → Midterm 1 on Friday, October 2 (exam not posted). **Week 7:** matrices → Ax → transforms + Quiz 5 (basis). **Quizzes and Labs 1–6 meet on Fridays** (Lab 0 is Week 1 only; no lab on Midterm 2 Friday or fall break).

Week 1

Mon Aug 24 — First day

No lecture outline. Class started Wednesday, August 26 with Lecture 01.

Wed Aug 26 — Why linear algebra? Vectors as representations

Lecture 01 page

Reading: Boyd 1.1

By the end of class, you should be able to:

1. Identify objects that can be represented as vectors.
2. Interpret vector entries, units, and indexing in context.
3. Distinguish scalars from vectors.
4. Translate a small data, position, signal, or measurement problem into vector form.

Fri Aug 28 — Vector arithmetic and geometry

Lecture 02 page

Reading: Boyd 1.2–1.3

By the end of class, you should be able to:

1. Add, subtract, and scale vectors.
2. Interpret those operations geometrically.
3. Plot vectors in $\mathbb{R}^2$ and recognize corresponding operations in $\mathbb{R}^3$.
4. Check whether an answer has sensible dimensions and units.

Week 2

Mon Aug 31 — Linear combinations

Lecture 03 page

Reading: Linear combinations (class notes), Lab 0, and Lachniet Ch. 1 (Getting Started).

No prior coding assumed. Lachniet is one PDF on Resources; chapters are listed here.

By the end of class, you should be able to:

1. Form and interpret linear combinations.

2. Explain the meaning of coefficients in context.
3. Enter, index, and modify vectors in Octave or Python.
4. Perform vector arithmetic and create a basic plot.
5. Modify supplied code and explain its output or error message.

Wed Sep 2 — Inner product

Lecture 04 page

Reading: Boyd 1.4; Boyd 1.1 stack and slice (Quiz 1 prep)

By the end of class, you should be able to:

1. Compute a dot product.
2. Interpret it as a weighted sum.
3. Interpret its sign and magnitude as alignment.
4. Recognize orthogonal vectors from a zero dot product.

Fri Sep 4 — Linear and affine functions

Lecture 05 page

Reading: Boyd 2.1 and 2.3

Boyd's calculus-dependent Taylor-approximation material is optional.

By the end of class, you should be able to:

1. Distinguish linear from affine functions in examples.
2. Test superposition numerically on simple functions.
3. Construct a function $a^\mathsf{T}x + b$ for a contextual quantity.
4. Evaluate a model by hand and computationally.

Week 3

Mon Sep 7 — Labor Day — no class

Wed Sep 9 — Linear functions in applications

Lecture 06 page

Reading: Applications of linear and affine models

By the end of class, you should be able to:

1. Model a score, cost, prediction, or aggregate quantity.
2. Interpret model coefficients and intercepts.
3. Compare outputs from competing models.
4. State a reasonable limitation or assumption of a linear model.

Fri Sep 11 — Norm, distance, RMS, and variation

Lecture 07 page

Reading: Boyd 3.1–3.3

By the end of class, you should be able to:

1. Compute vector norms and distances.
2. Interpret norm and distance geometrically and contextually.
3. Relate norm to RMS and, where appropriate, variation around an average.
4. Use software to calculate distances in dimensions too large for hand computation.

Week 4

Mon Sep 14 — Angle and similarity

Lecture 08 page

Reading: Boyd 3.4

By the end of class, you should be able to:

1. Compute cosine similarity or an angle between vectors.
2. Distinguish closeness in position from similarity in direction.
3. Choose an appropriate similarity measure for a stated purpose.
4. Explain why normalization may change a comparison.

Wed Sep 16 — Nearest neighbors

Lecture 09 page

Reading: Boyd §3.2 nearest neighbor; exercises 3.10–3.11, 3.24

Possible contexts: species measurements, geographic sites, songs, or sensor readings. (Boyd Ch. 4 begins clustering; NN is Chapter 3.)

By the end of class, you should be able to:

1. Formulate a nearest-neighbor question using vectors.
2. Find a nearest neighbor in a small dataset by hand.
3. Explain how feature scale can affect the answer.
4. Justify a choice of distance or similarity measure.

Fri Sep 18 — Clustering and k-means

Lecture 10 page

Reading: Boyd 4.1–4.3

Same day as Quiz 3.

By the end of class, you should be able to:

1. Explain the goal of clustering.
2. Compute one assignment/update iteration of k-means.
3. Interpret a centroid as a vector.
4. Recognize that clustering output depends on representation, scaling, and initialization.

Week 5

Mon Sep 21 — Linear dependence as redundancy

Lecture 12 page

Reading: Boyd 5.1

By the end of class, you should be able to:

1. Recognize simple dependent and independent collections.
2. Construct a dependence relationship.
3. Interpret dependence as redundant information.
4. Explain why redundancy matters in data, measurement, and computation.

Wed Sep 23 — Span, independence, and subspaces

Lecture 13 page

Reading: Traditional vocabulary supplement to Boyd

By the end of class, you should be able to:

1. Define span using linear combinations.
2. Determine whether a vector lies in a simple span.
3. Interpret spans geometrically as lines, planes, or higher-dimensional collections.
4. Recognize basic subspaces and distinguish them from shifted lines or planes.

Fri Sep 25 — Basis, dimension, and orthonormal vectors + Quiz 4

Lecture 14 page · Quiz 4

Reading: Boyd 5.2–5.3

Quiz 4 is Week 4 material (Lectures 08–10: angle, neighbors, clustering) — not basis.

By the end of class, you should be able to:

1. Explain what a basis accomplishes.
2. Identify a basis and dimension in simple examples.
3. Use linear combination, span, subspace, independence, basis, and dimension accurately.
4. Recognize an orthonormal collection.
5. Explain why orthonormal coordinates are convenient for measurement and reconstruction.

Week 6

Mon Sep 28 — Neighbors practice

Lecture 11 page · Syntax reference

Reading: Lachniet Ch. 1; Lectures 08–10. No new Boyd section.

Ungraded software practice on distances and nearest neighbors. Quiz 4 was Friday, September 25. Graded Lab 1 is Friday, October 16.

By the end of class, you should be able to:

1. Load and inspect a dataset represented by vectors.
2. Compute many distances with Octave or Python.
3. Adapt supplied code to identify nearest observations or clusters.
4. Validate computational output on a small hand-checkable subset.
5. Interpret results in the original context.

Wed Sep 30 — Midterm 1 prep

Lecture 15 page

Reading: Midterm 1 study guide and practice (no new Boyd section; no separate lecture notes)

Entire class is prep with those packets. Midterm 1 is Friday (Lecture 16); the exam itself is not posted online.

By the end of class, you should be able to:

1. Connect distance, similarity, span, and independence.
2. Select an appropriate vector tool for a new problem.
3. Move between algebraic, geometric, computational, and contextual representations.
4. Diagnose and explain common errors.
5. Identify gaps and make a targeted study plan before Friday.

Fri Oct 2 — Midterm 1

Lecture 16 page

Reading: Midterm 1 (in class; exam not posted)

Entire class period. Prep materials stay on Lecture 15.

By the end of class, you should be able to:

1. Compute with vectors and linear functions.
2. Interpret vector operations in context.
3. Analyze distance, similarity, span, and independence.
4. Select and communicate an appropriate method.

Week 7

Mon Oct 5 — Matrices as data and structured objects

Lecture 18 page

Reading: Boyd 6.1–6.3

By the end of class, you should be able to:

1. Identify matrix dimensions, rows, columns, and entries.
2. Represent datasets, networks, images, or systems as matrices.
3. Perform matrix addition, scaling, and transposition.
4. Recognize zero, identity, diagonal, triangular, symmetric, and sparse matrices.
5. Explain one practical consequence of matrix structure.

Wed Oct 7 — Matrix-vector multiplication

Lecture 19 page

Reading: Boyd 6.4

Connect directly back to Week 1 linear combinations.

By the end of class, you should be able to:

1. Compute Ax by hand for a small matrix.
2. Interpret Ax as a linear combination of the columns of A.
3. Interpret A as an input-output transformation.
4. Explain how the dimensions of A determine the input and output spaces.

Fri Oct 9 — Quiz 5; transforming space taught Oct 14–16

Lecture 20 page · Quiz 5

Reading: Boyd 7.1–7.2 and Lachniet Ch. 2 (matrices and linear-algebra computations).

Quiz 5 is basis (Lecture 14), not transforms. The transform lecture itself was taught October 14 and 16. It is on Quiz 7, which is online in Canvas on October 23. Lab 2 that day is Gaussian elimination.

By the end of class, you should be able to:

1. Predict the effect of scaling, reflection, shear, and rotation matrices.
2. Apply matrices to many 2D or 3D points at once.
3. Plot and compare original and transformed objects.
4. Adapt a short program that implements a transformation.
5. Interpret a matrix as a coordinate or physical-space transformation.

Week 8

Mon Oct 12 — No class

No meeting.

Wed Oct 14 — Transforming space, started

Continued Friday. Notes: Lecture 20. Objectives are on the Oct 9 Lecture 20 block.

Fri Oct 16 — Transforming space, finished + Quiz 6 + Lab 1

Lecture 20 · Quiz 6 · Lab 1

Quiz 6 is matrices and Ax (Lectures 18–19), not transforms and not Lab 1. Transforms are on Quiz 7.

Week 9

Mon Oct 19 — Systems, $Ax = b$, and elimination

Lecture 21 page

Reading: Boyd 8.3, plus Gaussian elimination (class notes) and Lachniet Ch. 2.

One meeting for the old systems lecture and the old elimination lecture.

By the end of class, you should be able to:

1. Translate equations into $Ax = b$ and classify the system as square, overdetermined, or underdetermined.
2. Interpret $Ax = b$ as asking whether b is in the span of the columns, and anticipate no solution, one solution, or infinitely many.
3. Match each elementary row operation with its matrix, use those matrices to eliminate, and solve a small system by back substitution.
4. Explain why those row operations preserve the solution set.

Wed Oct 21 — RREF and linear dynamical systems

Lecture 22 page

Reading: RREF (class notes), Lachniet Ch. 2, and Boyd 9.1–9.3.

One meeting for RREF and the first dynamics lecture. Iteration and long-run pattern names are Lecture 23 on Friday.

By the end of class, you should be able to:

1. Read a reduced row echelon form: name pivot and free variables and write a parametric solution.
2. Tell inconsistent, unique, and infinitely many solutions apart from that form.
3. Write an evolving system as $x_{t+1} = Ax_t$ and build the transition matrix from a flow description.
4. Compute and interpret the next state or two.

Week 10

Mon Oct 26 — Lab 2 / RREF catch-up

Continuation of Lab 2 / RREF practice as needed.

Wed Oct 28 — From row operations to matrix products

Lecture 23 page

Reading: Boyd 10.1; review RREF and one-step dynamics from Lecture 22.

Builds on elementary matrices (already used on augmented systems) and extends matrix–vector multiplication to matrix–matrix products, with a short dynamics bridge.

By the end of class, you should be able to:

1. Apply an elementary matrix to an augmented matrix and read the solution set (including a parametric form).
2. Classify unique / none / infinitely many from RREF and write infinite families with a parameter t.
3. Build a transition matrix from a two-compartment flow (rows = to, columns = from) and compute one step $x_1 = Ax_0$.

4. Explain why two steps need $A(Ax_0)$ / A^2, and compute a small product AB by hand.
5. Give an example with $AB \neq BA$, and compute A^2 and a small power $A^n x$.

Fri Oct 30 — Quiz 8 + Lab 3

Quiz 8 · Lab 3

Reading: Lab 3; optional Boyd complexity remarks.

Friday is **not** a numbered lecture. Quiz 8 is systems / elimination. Lab 3 carries cost / solve / powers syntax and a $[A \mid I]$ peek toward Wednesday's inverse.

By the end of Lab 3, you should be able to:

1. Distinguish storage, processing work (operation count), and wall-clock time.
2. Count ops on a small elimination and see the $O(n^3)$ pattern.
3. Enter A, b, solve, check residual, and compute matrix powers in Octave/NumPy.
4. Package elimination as $A = LU$ (first useful factorization; contrast with $[A \mid I]$ for A^{-1}).
5. Preview $[A \mid I]$ toward Wednesday's inverse.

Week 11

Mon Nov 2 — Lab 3 continued (not a numbered lecture)

Lab 3

More work on Lab 3 (cost / LU / solve / $[A \mid I]$ peek). No new lecture number today.

Wed Nov 4 — Inverses, rank, and solvability

Lecture 24 page

Reading: Boyd 11.1–11.3; class notes + selected Strang §3.5 (rank, $C(A)$, $N(A)$); Lab 3 peek.

Why in this course: Inverse = undo map when A is square and full rank. $C(A)$ / $N(A)$ / rank = solvability and uniqueness language for general A (including rectangular / singular). Not a four-subspaces theory unit.

By the end of class, you should be able to:

1. Explain why reducing $[A \mid I]$ to $[I \mid B]$ gives $B = A^{-1}$, and use an inverse to solve $Ax = b$.
2. Interpret $C(A)$ as possible outputs and $N(A)$ as inputs sent to zero.
3. Find rank from pivots and use $\dim N(A) = n - \text{rank}(A)$.
4. Explain when $Ax = b$ is solvable and when a solution is unique.
5. Distinguish square invertibility from full column rank for a tall matrix.

Fri Nov 6 — Determinants, geometry, and invertibility + Quiz 9 + Lab 4

Lecture 25 page · Quiz 9 · Lab 4

Reading: Determinants (class notes); Lab 4 practices inverses / rank / determinants in software.

Quiz 9 is Lecture 22 only (RREF / one-step dynamics), not today's determinants and not Lab 4.

By the end of class, you should be able to:

1. Compute determinants of 2×2 and simple triangular matrices.
2. Interpret determinant magnitude as area or volume scaling.
3. Connect determinant zero with lost dimension and noninvertibility.
4. Recognize the determinant of a triangular matrix from its diagonal.
5. Connect determinants to the coming characteristic equation.

Week 12

Mon Nov 9 — Midterm 2 review and synthesis

Lecture 26 page

Reading: Unit II review

Full review day.

By the end of class, you should be able to:

1. Connect transformations, systems, dynamics, rank, inverses, and determinants.
2. Compare hand and computational solution methods.
3. Interpret software output instead of merely reporting it.
4. Solve a new application requiring more than one concept.

Wed Nov 11 — Veterans Day — no class

Fri Nov 13 — Midterm 2

Lecture 27 page

Reading: Midterm 2 (in class)

By the end of class, you should be able to:

1. Analyze matrices as transformations and models.
2. Solve and interpret systems.
3. Use matrix products, powers, rank, inverses, and determinants appropriately.
4. Interpret computational results in context.

Week 13

Mon Nov 16 — Eigenvectors as persistent directions

Lecture 28 page

Reading: Beginning of Strang §6.1 — $Ax = \lambda x$, eigenvectors, and dynamics.

Arc: L28 name eigen / $A^t v = \lambda^t v$ (campus matrix preview) → L29 hand 2×2 + complex → L30 + Lab 5 discover the long-run mix by iteration, then explain ($\lambda = 1$ wins). The free PDF is only §6.1.

By the end of class, you should be able to:

1. Explain geometrically what makes v an eigenvector.
2. Verify a proposed eigenpair.
3. Find λ when an eigenvector is given.
4. Connect eigenvectors to repeated transformations and steady directions.
5. Recognize that a rotation may have no real eigenvector, motivating complex numbers.

Wed Nov 18 — Characteristic equation, 2×2 eigenproblems, and complex numbers

Lecture 29 page

Reading: Strang §6.1, characteristic equation and the complex-eigenvalue (90° rotation) example.

Meets the explicit hand-calculation and complex-number CLOs. The same free §6.1 file is used Nov 16–20.

By the end of class, you should be able to:

1. Interpret $a + bi$ algebraically and as a point/vector in the complex plane.
2. Perform the basic complex arithmetic needed for eigenvalue calculations.
3. Form and solve $\det(A - \lambda I) = 0$ for a 2×2 matrix.
4. Find corresponding eigenvectors.
5. Check real or complex eigenpairs by substitution.

Fri Nov 20 — Why did the system behave that way? + Quiz 10 + Lab 5

Lecture 30 page · Quiz 10 · Lab 5

Reading: Strang §6.1, Lab 5, and Lachniet Ch. 5 (eigenvalues/eigenvectors).

Normal modes appear as an application demo (not a differential-equations course). Quiz 10 is Week 11 (inverses / rank / determinants), not today's eigen lab.

By the end of class, you should be able to:

1. Compute eigenvalues and eigenvectors of a general matrix with software.
2. Verify $Av \approx \lambda v$ numerically.
3. Use dominant eigenbehavior to explain a simulated system.
4. Interpret complex eigenvalues as rotation/oscillation behavior.
5. Explore a symmetric-matrix or normal-mode example and observe orthogonal eigenvectors.

Week 15

Mon Nov 30 — When $Ax = b$ has no exact solution

Lecture 31 page

Reading: Boyd 12.1

By the end of class, you should be able to:

1. Recognize an inconsistent overdetermined system.
2. Formulate the least-squares objective $\min_x \|Ax - b\|$.
3. Interpret $\hat{b} = A\hat{x}$ as the orthogonal projection of b onto $C(A)$.
4. Interpret the residual as orthogonal to $C(A)$ at a minimizer.
5. Distinguish an exact solution from a best approximate solution.

Wed Dec 2 — Least squares, data fitting, and positive definiteness

Lecture 32 page

Reading: Boyd 12.2–12.4 and 13.1, class notes on symmetric and positive-definite matrices, and Lachniet Ch. 2 (least-squares / data-fitting exercises).

Positive definiteness is from Boyd and class notes. Friday is project launch with Lecture 33 (SVD) and Lab 6 (Fit and Compress); Quiz 11 is Week 13 (eigenvalues), not least squares.

By the end of class, you should be able to:

1. Compute a least-squares solution with Octave or Python.
2. Plot observed and fitted values.
3. Interpret residuals and an RMS error in context.
4. Check the residual's orthogonality to the columns of A.
5. Evaluate whether a fitted model is useful rather than merely reporting coefficients.
6. See basis choice (monomials vs truncated powers) as choosing columns of A / the model subspace.
7. Recognize that $A^\mathsf{T}A$ is symmetric positive semidefinite, explain $x^\mathsf{T}A^\mathsf{T}Ax = \|Ax\|^2$, and recognize that $A^\mathsf{T}A$ is positive definite when A has linearly independent columns.

Fri Dec 4 — SVD, Quiz 11, and project launch

Lecture 33 page · Quiz 11 · Capstone / project launch

Reading: Selected slides from the Strang/MIT SVD slides (not all 17 pages), capstone launch, Lachniet Ch. 5 (SVD) and Ch. 7 (image compression). Same-day software: Lab 6 (Fit and Compress — LS + truncated SVD).

Quiz 11 is Week 13 eigenvalues, not today's SVD. The period is quiz + **SVD** + Lab 6 + **project launch**; the day title is project launch.

By the end of class, you should be able to:

1. Interpret $A = U\Sigma V^\mathsf{T}$ conceptually.
2. Construct a rank-k approximation computationally.
3. Compare singular-value decay for an image/frame and a frame-difference matrix.
4. Encode three frames either independently or as F_1, D_2, D_3.
5. Compare reconstruction error and storage for the two approaches.
6. Formulate a capstone question involving accuracy, storage, runtime, and resource use.

Week 16

Mon Dec 7 — Capstone studio

Lecture 34 page

Reading: Capstone (no new mathematics)

By the end of class, you should be able to:

1. Implement and validate a low-rank computational strategy.
2. Measure approximation error, storage, and runtime.
3. Separate one-time factorization cost from repeated-use cost.
4. Compare theoretical operation/storage counts with empirical measurements.
5. Debug and refine an analysis collaboratively while retaining individual understanding.

Wed Dec 9 — Capstone presentations / gallery walk

Lecture 35 page

Reading: Capstone presentations

Pair lightning talks or gallery walk.

By the end of class, you should be able to:

1. Explain the problem and why SVD or another linear-algebra tool was appropriate.
2. Present quantitative evidence about accuracy and cost.
3. Defend a choice of rank or representation.
4. Evaluate classmates' methods constructively.
5. Communicate the practical value and limitations of the result.

Fri Dec 11 — Cumulative final review

Lecture 36 page

Reading: Cumulative review

No routine quiz or new content.

By the end of class, you should be able to:

1. Connect vectors, systems, transformations, dynamics, eigenvectors, least squares, and SVD.
2. Select an appropriate method for a new situation.
3. Decide what should be done by hand and what should be delegated to software.
4. Interpret mathematical and computational results.
5. Create a targeted final-exam study plan.